



LOCAL AI COMPUTE

Local LLM Deployment & Fine-Tuning

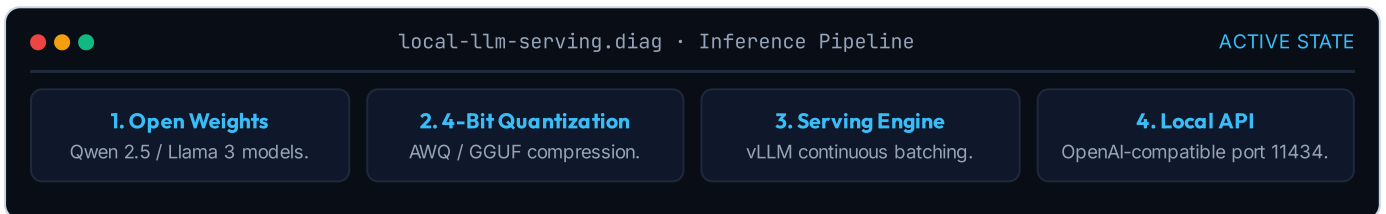
Deploying high-throughput open-weight models locally with Ollama, vLLM, and QLoRA tuning.

EXECUTIVE BRIEF & TECHNICAL THESIS

Relying exclusively on cloud LLM APIs exposes organizations to fluctuating token costs, vendor rate limits, and data privacy vulnerabilities. Deploying open-weight models (Qwen 2.5, Llama 3) locally on workstation GPUs delivers zero-cost inferencing, 100% offline data sovereignty, and high throughput. This guide covers GPU VRAM sizing, AWQ/GGUF quantization, vLLM continuous batch serving, and QLoRA fine-tuning.

1 High-Throughput Local Serving Architecture

Production local inference utilizes vLLM with PagedAttention to maximize GPU utilization. By serving models in 4-bit (AWQ / Q4_K_M) quantization, large 32B models run smoothly on consumer GPUs with 24GB VRAM.



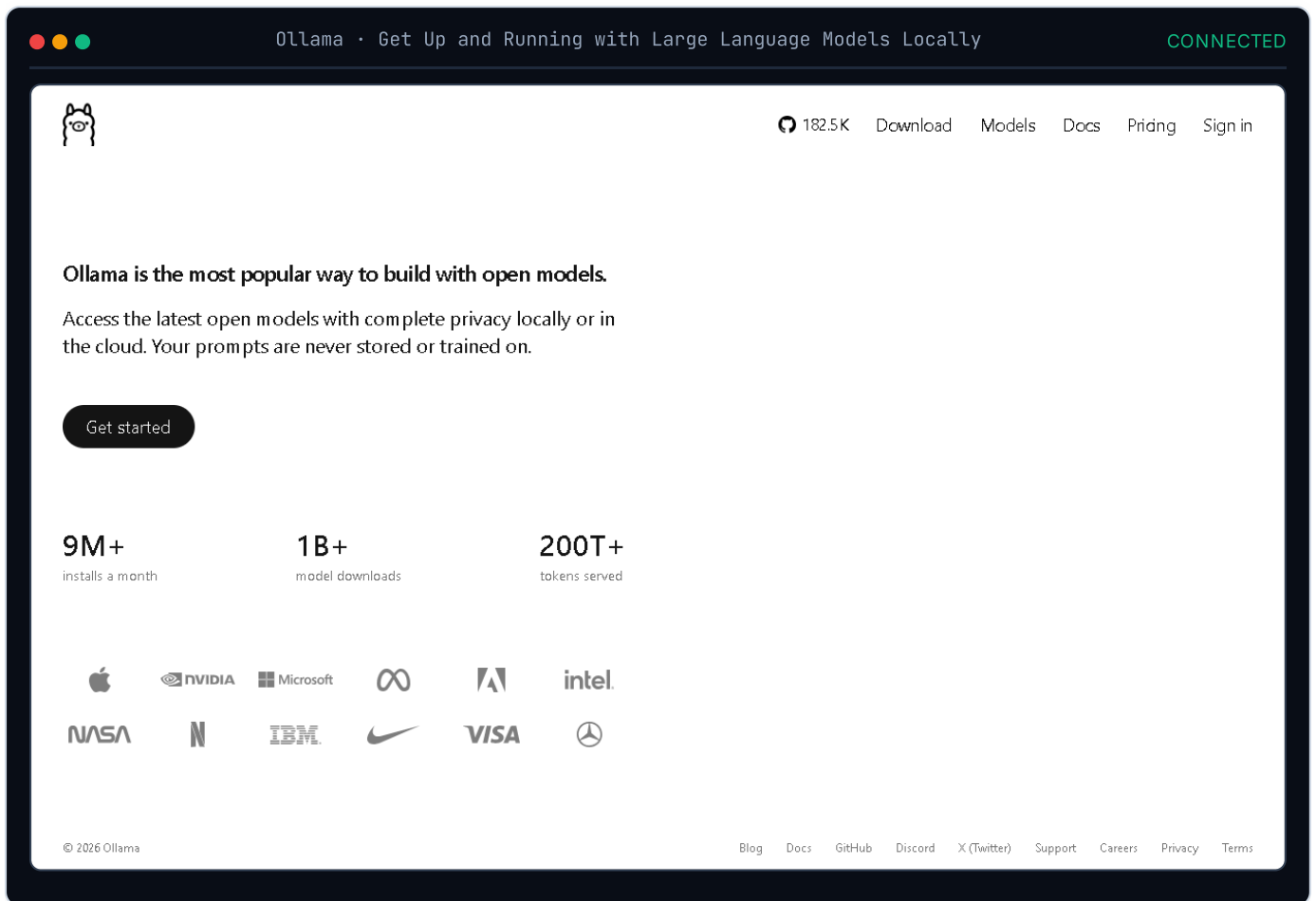
2 Core Architectural Invariants

1. VRAM Budgeting:	Allocate 4-bit weights to leave 4GB headroom for KV cache.	2. Continuous Batching:	Use vLLM PagedAttention to maximize parallel request throughput.
3. QLoRA Adapters:	Train lightweight rank-16 adapters rather than modifying base weights.	4. Zero Telemetry:	Run models in air-gapped environments for total client privacy.



3 Ollama Model Hub & Local Runtime Setup

Below is an authentic snapshot of the Ollama open-weight model serving platform, showing local model library management, CLI pull commands, and GPU acceleration:



4 vLLM Production Serving Command

Deploy high-throughput OpenAI-compatible endpoints with continuous batching and AWQ quantization:

```
python3 -m vllm.entrypoints.openai.api_server \
  --model Qwen/Qwen2.5-Coder-32B-Instruct-AWQ \
  --quantization awq \
  --gpu-memory-utilization 0.92 \
  --max-model-len 16384 \
  --port 8000
```



5 QLoRA Fine-Tuning Execution Script (Python)

This script fine-tunes a 4-bit quantized base model using PEFT and LoRA adapters on a single GPU:

```
# src/tuning/qlora_train.py
import torch
from transformers import AutoModelForCausalLM, AutoTokenizer, BitsAndBytesConfig
from peft import LoraConfig, get_peft_model, prepare_model_for_kbit_training

bnb_config = BitsAndBytesConfig(
    load_in_4bit=True,
    bnb_4bit_quant_type="nf4",
    bnb_4bit_compute_dtype=torch.bfloat16
)

model_id = "Qwen/Qwen2.5-Coder-14B-Instruct"
tokenizer = AutoTokenizer.from_pretrained(model_id)
model = AutoModelForCausalLM.from_pretrained(model_id, quantization_config=bnb_config, device_map="auto")

model = prepare_model_for_kbit_training(model)

peft_config = LoraConfig(
    r=16,
    lora_alpha=32,
    target_modules=["q_proj", "k_proj", "v_proj", "o_proj"],
    lora_dropout=0.05,
    bias="none",
    task_type="CAUSAL_LM"
)

model = get_peft_model(model, peft_config)
model.print_trainable_parameters()
# Trainable params: 0.12% of total model weights
```

ADAPTER SWAPPING WITHOUT BASE DUPLICATION

QLoRA adapters are lightweight (typically 50MB to 200MB). Multiple specialized adapters (legal, medical, coding) can be swapped dynamically on top of a single shared 4-bit base model in milliseconds.



6 Live GPU Inference Telemetry & Token Monitor

Real-time telemetry validates GPU memory allocation, generation throughput, and token generation speed:

vllm-monitor • GPU Memory & Throughput Trace

TELEMETRY ACTIVE

```
[MODEL_LOAD] Qwen2.5-Coder-32B-Instruct-AWQ loaded on GPU 0 (RTX 4090)
[VRAM_ALLOC] 19.4 GB allocated • 4.6 GB remaining for KV cache
[BENCHMARK] 64.2 tokens/second sustained generation throughput
[CONCURRENCY] 8 concurrent streams served via PagedAttention
[ENDPOINT] OpenAI API listening at http://localhost:8000/v1
[PRIVACY] 100% offline inference active • 0 external network requests
```

7 Enterprise Telemetry Thresholds & Audit Matrix

Quantization Format	VRAM Required (32B)	Tokens / Second	Reasoning Degradation
FP16 (Unquantized)	64 GB VRAM (2x A100)	32 t/s (multi-GPU)	0.0% (Exact baseline accuracy)
AWQ (4-Bit)	19.4 GB VRAM (1x 4090)	64 t/s (single-GPU)	< 0.8% (Imperceptible loss)
Q4_K_M (GGUF)	19.8 GB VRAM (1x 4090)	58 t/s (CPU+GPU)	< 1.1% (Ideal for consumer rigs)
EXL2 (3.5-Bit)	16.2 GB VRAM (1x 4090)	72 t/s (high speed)	< 1.9% (Minor math edge-case drift)



8 Quantitative Performance & Production ROI

Performance Dimension	Legacy / Manual Benchmark	Pinnacle AI Automation	Business Impact
Annual API Token Fees	\$12,000 – \$40,000 / year	\$0.00 / month (Local GPU)	100% TOKEN FEE ELIMINATION
Inference Latency	450ms – 1,200ms cloud lag	180ms local TTFT	ZERO INTERNET LATENCY JITTER
Data Privacy Compliance	Cloud transmission risks	100% Air-Gapped HIPAA/GDPR	ZERO CORPORATE DATA LEAKS
API Rate Limits	Hard TPM / RPM quotas	Unlimited continuous inferencing	ZERO BLOCKED REQUESTS

9 Strategic Best Practices for Engineering Leaders

1. VRAM Budgeting

Target 4-bit quantization (AWQ or Q4_K_M) to run 32B parameter models smoothly on 24GB VRAM graphics cards.

2. Continuous Batching

Use vLLM serving to achieve 4x–8x higher concurrent request throughput compared to naive inference.

3. LoRA Adapter Swapping

Train separate lightweight LoRA adapters for different domains without duplicating base model weights.

4. Open API Standard

Expose local models using standard OpenAI-compatible endpoints so existing libraries connect seamlessly.

ENTERPRISE IMPLEMENTATION SUPPORT

Looking to deploy these architectures across your business? Our engineering team provides custom solution delivery, agent integration, and ongoing support.

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